Digital Archeology: Features and Metrics to Quantify the Degree of Changes in Digital Online Projects

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Research in the digital humanities is usually connected to an online project. These projects make research accessible and public facing, but susceptible to unintended and unexpected changes due to various factors. Thus, the degradation of online projects is a complex problem, making it understated and misunderstood. It is not a problem that involves just two states, but one that gradually occurs.

Fitzpatrick (Fitzpatrick, 2011) focuses on the technological changes (notably greater utilization of Internet publication technologies and digital archiving) that are necessary to allow academic publishing to thrive into the future, highlighting that the key issues that must be addressed are social and institutional in origin. Studies have been carried out to address potential reconstruction (Klein et al., 2011), the overall decay of websites (Bar-Yossef et al., 2004), institutional longevity (VandeCreek, 2022), and the decomposition of shared resources (SalahEldeen and Nelson, 2012). While recommendations are in place for creating stable and long-lasting resources (Arneil et al., 2019), they can be difficult and problematic to implement for projects that have been running for some years.

Our previous work has incorporated machine learning and data mining to address digital preservation questions, focusing on identifying and managing the signs of degradation of online projects in the digital humanities (Meneses et al., 2020), and examined cyber threats as a significant factor in their decay (Meneses and Martin, 2023). For this abstract, we focus on an exploration of features: the individual salient measurable properties or characteristic of signs of degradation in online projects, that we intend to parse and analyze with machine learning (in an upcoming continuation to this study) as an effort to lay the groundwork for developing metrics to quantify the degree of changes in online projects over time. Our purpose is to provide systematic methods to sustain and preserve culture and the digital scholarly infrastructure in the humanities over time, preventing their degradation and decay.

For this purpose, we have been analyzing projects that have been published in the DH Book of Abstracts (Karajgikar et al., 2024) for the last 17 years (2007 to 2024), analyzing their changes and shifts over time. We start by storing periodic snapshots for each project as WARC files (International Organization for Standardization, 2017), examining their functioning and non-functioning components, and collecting the text from these documents using Python's BeautifulSoup Library (Richardson, 2015). We also use Python's Requests (Reitz, 2019) to retrieve the HTTP response codes and headers, which will be used to classify the snapshots by their year of publication into groups depending on the status of its server response. Through this process, focused on feature extraction, we can track the changes of these projects from their published date to their potential points of abandonment.

The features we have identified fall into two categories. First, explicitly observable features, which are features associated with the visually observable characteristics of the project; and second, implicit features which are potentially external to the project site itself. Our hypothesis is that implicit features are more revealing of the long-term health of the project's infrastructure, whereas the explicit features would be an acute sudden change that points towards a more implicit cause. For example, some projects can show signs of degradation because of lack of financial resources, or the project changing hands. These features can be difficult to capture explicitly from analyzing a project, but directly impact its long-term health. Alas, one could think of the explicit/observable features as being symptomatic of the underlying cause. A broken hyperlink/image may be associated with a server change-over, or a redistribution of assets. The explicit features could be seen as a sign of alarm for a project potentially undergoing degradation.

For the upcoming continuation of this study, we plan to analyze the features we collected and extracted using: Scikit-Learn's Logistic Regression and Support Vector Classification (scikit-learn, 2024), Transformers (Kamath et al., 2022), and

OpenCV (OpenCV, 2019). We must also emphasize that we are collecting data ethically with academic purposes: respecting authorship, exclusion protocols, and ensuring no redistribution. Our focus is solely on furthering the conversation on the preservation of knowledge.

Our study has allowed us to conclude 2 points. First, identifying and collecting features that will be analyzed with machine learning is feasible, and can be utilized to develop metrics to preserve knowledge in the digital humanities. Second, the methodology that we have described in our study can be applied to other online projects without institutional support that are at risk of disappearing, for example research on minorities and underrepresented groups. Therefore, our study aims to provide insights and preservation strategies that could be applied to other similar online projects in the digital humanities, as steps towards democratizing knowledge and preventing their degradation and decay.

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